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MARSBASE, AN EDUCATIONAL SIMULATION GAME

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ABSTRACT

An educational game simulating the first developments of a Martian base is presented. The user faces a satellite image of Mars and commands the astronauts. They can walk on the surface, drive a rover, extract ores, transform hydrogen and carbon dioxide in propellant, build a greenhouse, do farming, and so on. A user-friendly interface has been developed. The simulation looks like other well-known simulation games, as for example Age of Empires or Simcity. In its current version, the game is playable but only a few scenarios are available.

1. INTRODUCTION

Is it really possible to build a base on Mars and to achieve self-sufficiency? This problem has been addressed by several researchers, among them Zubrin in his plan to settle the red planet [1], and science fiction writers such as Robinson [2]. In order to get a better understanding of the basic requirements of such a project, a user-friendly simulator has been developed. However, the aim is not to propose a technical study. It is instead an educational game showing the main technological breakthroughs allowing long stays on the surface of Mars.

The first draft of the project has been presented in the previous Mars Society convention. A number of modifications have been made and new concepts have been implemented. The aim of this paper is to present the main parts of the game. It is divided into three parts. In the first one, we present the first panel of the game that makes it possible to choose all objects being sent to Mars. In the second part, we present the different actions that are available to the user. In the last part, we discuss the different scenarios and the perspectives of the game.

2. LIFT-OFF

In the first stage, the user has to choose between a prepared mission and an undefined mission. In the first case, the user immediately faces the Martian surface with a base already built. In the second case, he has to choose the objects to be carried by the rocket for a flight to the red planet. Oxygen, water, the number of astronauts, the power systems, the rovers and the habitat are important objects, but many others are proposed, such as fertilizer, various chemical elements, a greenhouse, a chemical unit, a factory, and so on. The number of days in space and on the planet and the maximum payload are important parameters of this initial stage. It is also possible to select the landing area. A database of Mars images proposed by NASA on Internet has been integrated in the game to obtain a more realistic simulation. Since it is an educational game, we have tried to respect some basic requirements, based on technical reports [3], [4]. If the user does not select enough oxygen, water, or food according to the number of astronauts and the number of days of the mission, a message is displayed to inform the user about the missing elements and the mission has to be redesigned. If the

maximum payload is reached, no more object can be added. However, the proposed limit is also variable in order to make this initial stage very flexible. It is important to note that all objects are parameterized in definition files, so that it is possible to change object features, to delete some of them or to add new ones independently from the programmer. Objects are split into 4 categories: chemical elements, industrial objects, buildings and complex objects that can not be produced from Martian resources. An image of the initial stage is presented in Figure 1.

3. SURVIVING AND GROWING

Once a mission has been designed, a new screen is proposed. The user faces a high-resolution satellite image of Mars and the buildings imported from the Earth. Astronauts and rovers can be moved easily with the mouse all around the map. Each object moves relatively quickly and regularly towards its destination, avoiding the others. The whole map is often bigger than the screen. All parts can be easily reached by moving the cursor of the mouse to the borders.

There is also a fix panel devoted to the interface with the user. This panel is divided into four parts, a small image of the entire region at the top, the information panel, the astronaut selection area and the action panel:

- The simulated terrain is bigger than the screen. A small image of the entire region is thus displayed at the top of the panel to facilitate the localization. Ores and ice are roughly indicated on this image.
- The second one provides information to the user, in particular the amount of ores, chemical elements and industrial objects that are available at the base or in a rover. Information on the life support system, oxygen, water, food and power is always visible. A blackboard is also proposed to inform the user of a particular event.
- The astronaut selection area enables the user to select an astronaut at the base, in a rover or in EVA (extravehicular activity), provided that the corresponding building or vehicle has been previously selected. A double click on the astronaut makes him going out in EVA.
- All actions, typically the different tasks that can be performed by the astronauts are embedded in the "action panel". A selected astronaut can carry or put down an object, start or stop a transformation (chemical or industrial or farming) or start the construction of a new building. Most actions are simply represented by a small picture. A click on it starts the process. Then the picture is slowly repainted until the process is achieved. Another click may stop the task before end.

The purpose of this simulation game is to show the basic principles that will make it possible to achieve self-sufficiency during a long stay on the surface of Mars. We therefore propose typical actions such as propellant production from hydrogen and carbon dioxide or silicon extraction from silicates (to build solar panels later on). Water can be produced from ice or it can also be extracted from the soil (in small amounts) using well known heating and then condensation techniques. Farming is possible under a greenhouse if some fertilizer is available. Complex tasks require complex equipment. To perform them (for instance silicon extraction from silicates), a chemical unit, a solar furnace or a factory can be imported from the Earth. All actions proposed in the action panel are defined in declaration files in order to make them independent from the programmer and to allow a flexible management. Each definition requires:

- A name (with the corresponding picture)

- A list of resources (other objects, for instance hydrogen) and eventually each amount.
- A list of products (other objects, for instance propellant) and eventually each amount.
- A list of tools (other objects, for instance a chemical unit).
- The power consumption.
- The time of the process.

The first objective of the user is to keep the astronauts alive during the stay on the surface. First of all, oxygen, water and food should be carefully maintained at the base. An astronaut out in EVA has also a limited oxygen amount. He should go back to the rover or to the habitat in a brief delay. A rover is also limited by the amount of propellant, it should not be sent too far. We believe that these constraints are important for the simulation because they impose the user to care first about surviving. They also suggest typical development strategies for the evolution of the Martian base. For instance, propellant should be produced prior to any important exploration for seeking silicates or hematite. An image of the game is presented in Figure 2.

4. CONCLUSION

The game is already playable and can be downloaded from the web [5]. However, there are many small details that remain to be implemented or modified. There is actually no sound or music and only 3 simple scenarios are available. In the first one, the user has to look for ore and extract it. In the second, the user has to build a greenhouse in order to produce food. In the third scenario, propellant has to be produced from hydrogen and carbon dioxide before fuelling the rovers. These scenarios show the basic tasks that will be performed on Mars in the first missions.

The perspectives are numerous:

- In the current version available on the web, memory constraints impose the same image of Mars in all scenarios, no sound and no video. The next version will probably require a different support.
- Objects and actions have not been carefully defined. Since they are independent from the programmer, appropriate declarations will be made later. Important changes may be considered.
- We are intending to develop "meta-actions" in order to integrate several actions in a single one. For instance, it would be interesting to extract a specific ore, carry, walk, put down and come back by a single click on a particular meta-action icon. The carry and put down buttons may become useless.
- After the first missions, humans should stay on the surface much longer. Each two years (game time), the simulator should propose a new flight to Mars (eventually unmanned) with a small payload. The user will have to expand the base while preserving life conditions. Objects manufactured on the red planet should progressively replace all complex objects imported from the Earth.

Many other interesting features have to be implemented, it would be too long to list them here. Finally, it is our hope that the Marsbase simulation game will soon become attractive. Then it could be used as an educational tool for an introduction to some of the basic problems of Martian bases.

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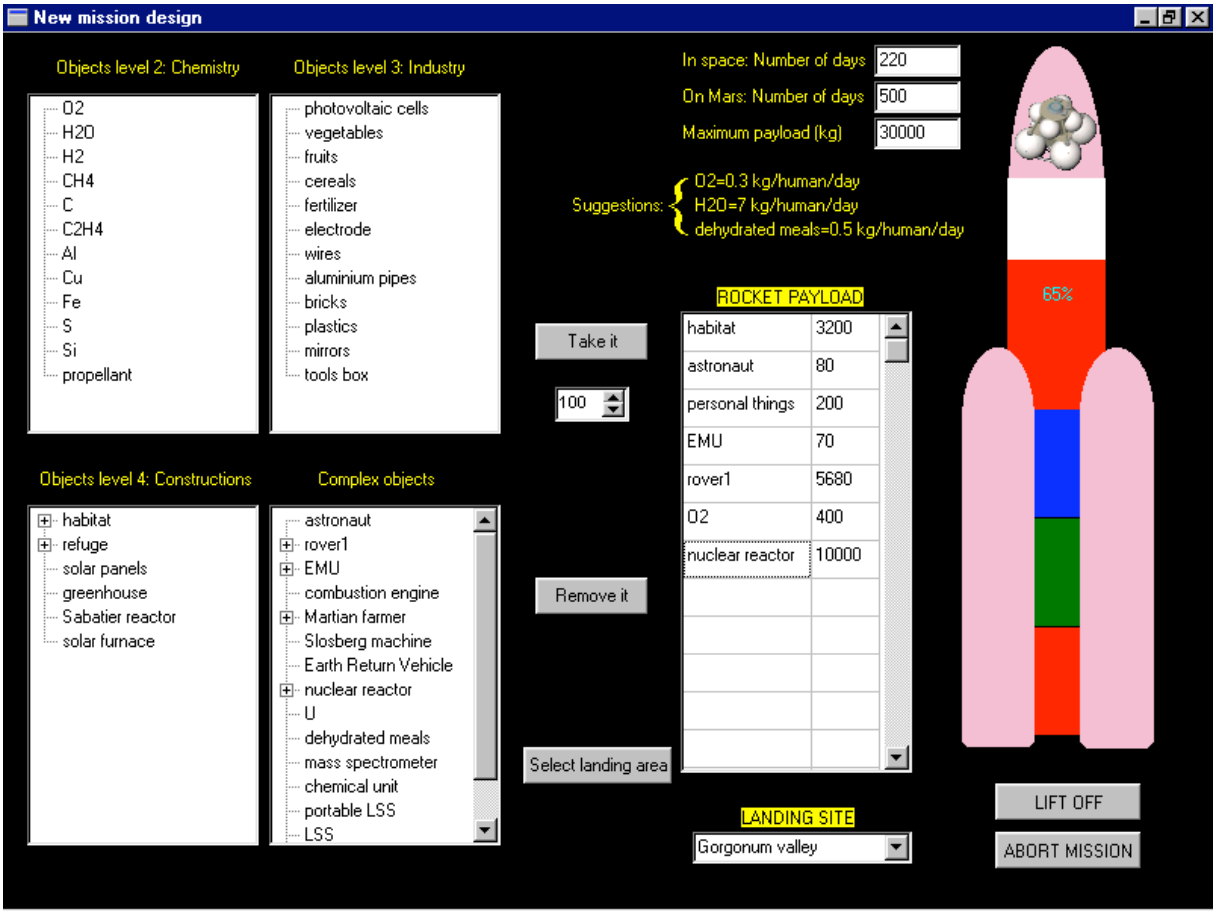

Figure 1: Initial stage, all objects to be carried by the rocket have to be chosen.

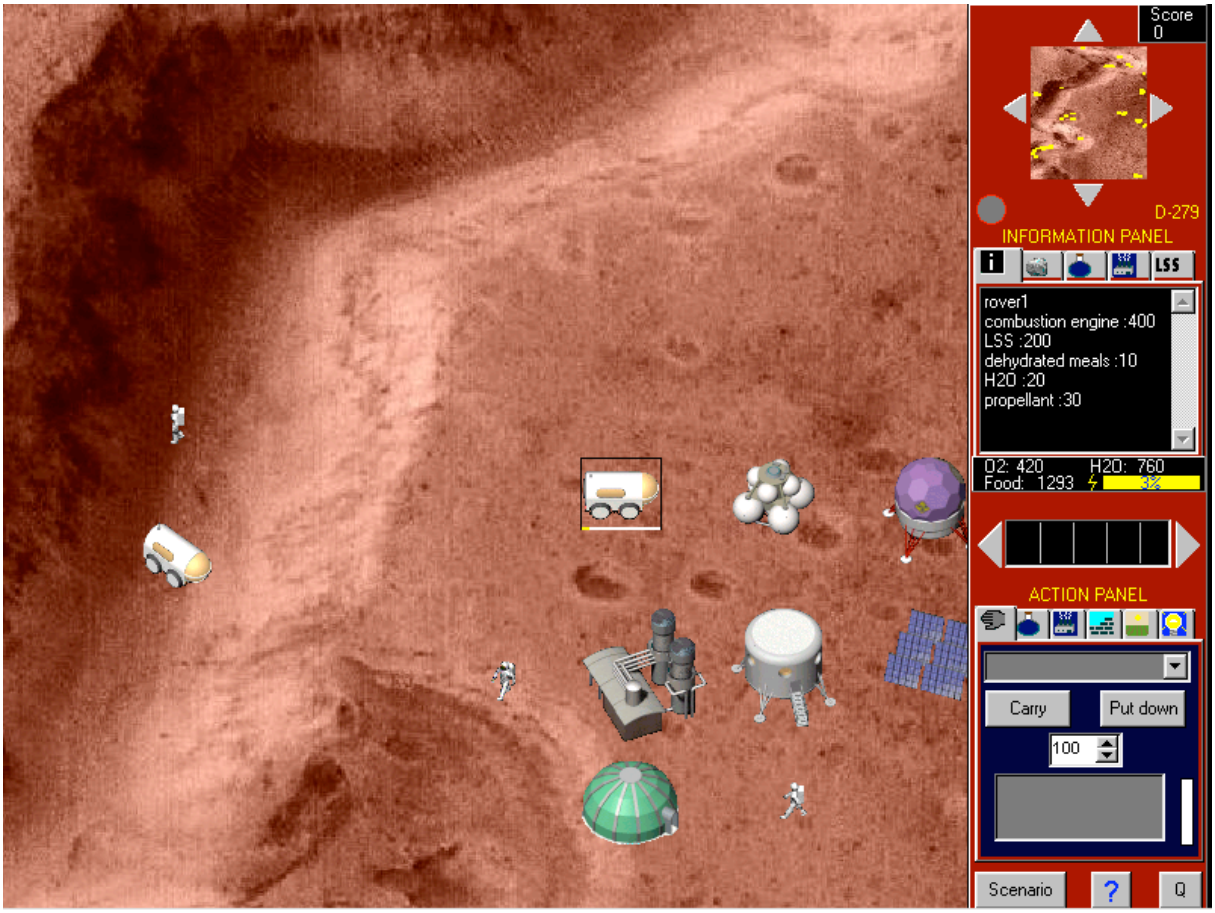

Figure 2: Image of the game. (Courtesy of NASA/JPL/Malin Space Science Systems)